



CORE PYTHON PROGRAMMING

Beginner to Advanced Level

Curriculum

MODULE 01 – Introduction to Python

- What is Python?
- Features & Applications
- Python Installation & IDE Setup
- Variables, Keywords & Identifiers
- Input & Output Functions

MODULE 02 – Data Types & Operators

- Numeric Data Types
- Strings & Boolean
- Type Conversion
- Arithmetic & Logical Operators
- Membership & Identity Operators

MODULE 03 – Strings in Python

- String Creation
- Indexing & Slicing
- String Methods
- String Formatting
- Practice Programs

MODULE 04 – Conditional Statements

- if, if-else, nested if
- elif Ladder
- match-case
- Real-Time Programs

MODULE 05 – Loops & Iterations

- while Loop
- for Loop & range()
- break, continue, pass
- Pattern Programs
- Fibonacci & Prime Numbers

MODULE 06 – Collections

- Lists & Comprehensions
- Tuples
- Sets
- Dictionaries
- Nested Collections

MODULE 07 – Functions

- Function Creation
- Parameters & Arguments
- Return Values
- Lambda Functions
- Recursion

MODULE 08 – Modules & Packages

- Import Statements
- Built-in Modules
- Custom Modules
- Packages & pip

MODULE 09 – Exception Handling

- Errors vs Exceptions
- try-except
- finally & else
- Custom Exceptions

MODULE 10 – File Handling

- File Operations
- Read & Write Files
- File Modes
- CSV Basics

MODULE 11 – Object Oriented Programming

- Classes & Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation & Abstraction

MODULE 12 – Advanced Python

- Iterators
- Generators
- Decorators
- *args & **kwargs

MODULE 13 – Database Connectivity

- SQLite Basics
- CRUD Operations
- Database Mini Project

MODULE 14 – Real-World Python

- Virtual Environments
- Project Structure
- API Introduction
- Python Career Paths

Capstone Project & Assessment

- ✓ Mini Projects
- ✓ Coding Exercises
- ✓ Assignments
- ✓ Final Project Presentation

"Master Python from Fundamentals to Real-World Applications"